

Object:- Write a program to find largest & smallest number from a given array.

Apparatus Required:- 5 Volt SMPS, Microprocessor 8085 Dyna Kit.

Steps for finding the large & smaller number

1. Initialize memory pointer & initialize counter c
2. Read number for the list store in B
3. If A is greater than B
4. Store large number in A
5. Increment the counter
6. Decrement counter c
7. If A is greater than B
8. Display the highest number.
9. Read number from the list

Input

F500 = 04

F501 = 07

F502 = 03

F503 = 0A

Output

F601 = 0A

Program - for finding ~~smallest~~ largest number

LXI H F500

MOV C, M

INX H

MOV A, M

DCR C

INX H

MOV B, M

CMP B

JNC F00E

MOV A, B

DCR C

JNZ F00F

STA F601

HLT

Teacher's Signature _____

input Values

F500 = 04

F501 = 07

F502 = 03

F503 = 0A

output

F601 = 03.

Expt. No. _____

Date _____

Page No. 04

for Adding largest number. smallest

LXI H, F500

MOV C, M

INX H

MOV A, M

INX H

~~MOV A,~~

DCR C

INX H

MOV B, M

CMP B

JC F00E

MOV A, B

DCR C

JNZ F007

STA F601

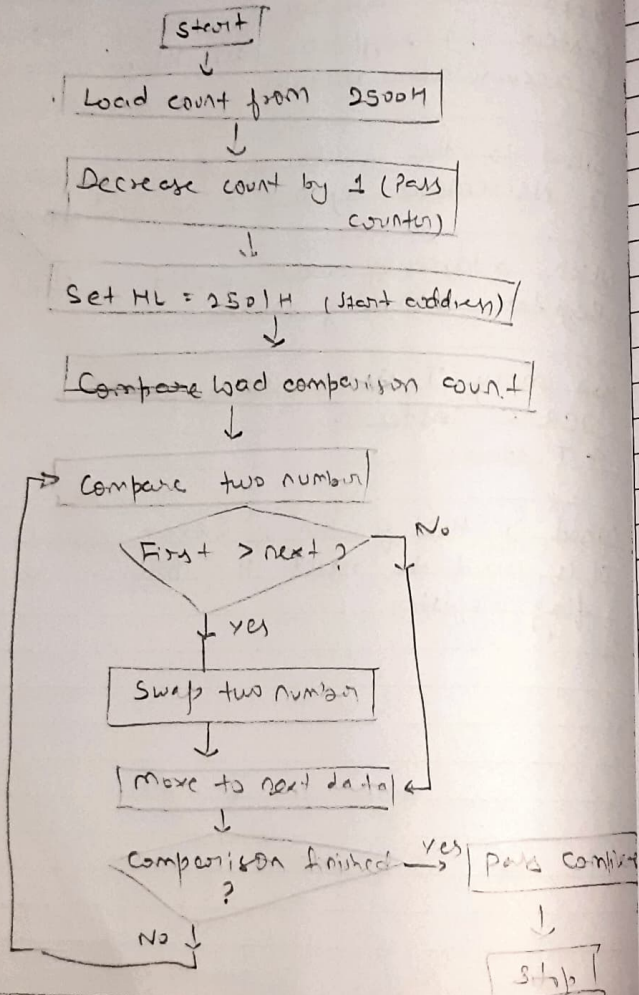
HLT

Teacher's Signature _____

Viva Questions

- Q1 What is the function of CMP M instruction?
Ans Compare the contents of memory ~~also~~ location pointed by register pair HL with the contents of accumulator results is not used.
- Q2 What is the function of DCR r instruction.
Ans To decrement by one in general purpose register.
- Q3 Which addressing mode is used in DAD r_s instruction?
Ans Register addressing mode.
- Q4 How many T states are required to complete DCR C instruction?
Ans 4 T states.
- Q5 What is the function of RESET signal?
Ans It is used to reset the program counter & flag register.

Algorithm flow



Expt. No. 9.

Date

Page No. 25.

Object:- Write a program to arrange a set (A) of data in ascending order.

Apparatus required:- +5 Volt, SNPS, Microprocessor 8085 Dyna-Kit.

Program Analysis- Search time for a sorted list of number is smaller than that of an unsorted list. There are many sorting algorithms available in the field of computer programming. e.g. Selection sort, merge sort, insertion sort, bubble sort etc. Bubble sort is used in this program. To sort the list in ascending order, we compare a number with the next number. If the later number is smaller than the first number then swap the number. This process is repeated with each other two adjacent number in list. In one pass, the largest number will come at the end of the list. In second pass, we repeat the process except the last position number & so on.

Teacher's Signature

Algorithm:-

1. Load the count of numbers from memory location 2500H
2. Decrease the count by 1
 This gives the number of comparison required in one pass.
3. Initialise pointer register pair (HL) to point to the first data element (2501H)
4. Compare the first element with the next element.
5. If the first number is greater than the second, swap them.
6. Move the pointer to the next pair of element.
7. Repeat steps 4 to 6 until one complete pass is finished.
8. After one pass $\rightarrow$ Decrease pass counter 1
9. Repeat the complete process until all passes are completed.

Program:- (Memorics)

```
LXI H, C050H
MOV C, H
DCR C
REPEAT LXI H, C050H
      MOV D, C
LOOP  MOV A, M
      INX H
      MOV B, M
      DCX H
```

Teacher's Signature _____

Observation

Input

C050 = Total no required = 5

C051 = 12

C052 = 10

C053 = 11

C054 = 8

C055 = 7

Output

C051 = 7

C052 = 8

C053 = 10

C054 = 11

C055 = 12

Expt. No. _____

Date _____

Page No. 27

	For Descending
CMP B	LXI H, C050H
JC NOSWAP	MOV C, M
MOX M, B	DEC C
INX H	Repeat LXI H, C050H
MOV M, A	MOV D, C
DCR D	Loop MOV A, M
JNZ LOOP	INX H
JMP COUNT	MOV B, M
NOSWAP INX H	DCX H
DCR D	CMP B
JNZ LOOP	JNC NOSWAP
COUNT DCR C	MOV M, B
JNZ REPEAT	INX H
HLT	MOV M, A
	DCR D
	JNZ LOOP
	JMP COUNT
	NOSWAP INX H
	DCR D
	JNZ LOOP

Result:- We have successfully performed to array of given number in ascending order.

Teacher's Signature _____

Viva Question

Q1 How many T-states are required to complete RRC instruction.

Ans 1 ϕ T state required to complete RRC instruction

Q2 How many machine cycles are required to complete RRC instruction?

Ans 1 machine cycle is required to complete RRC

Q3 Which addressing mode is used in HLT Command?

Ans Implicit addressing mode

Q4 What is the function of IN 8 bit port address?

Ans It is used to read a data byte from an input device.

Q5 Which addressing mode is used in IN & OUT?

Ans Implicit addressing mode.

Object:- Write a program to multiply 8 bit number

Apparatus:- +5Volts SMPS, Microprocessor 8085
Dyna Kit

Program Analysis

Algorithm

1. Load the first number from memory location 2500H
2. Store it in register B.
3. Load the second number from memory location 2501H
4. Store it in register C.
5. Clear accumulator A
- This will store the lower byte to result
6. Clear register D
- This will store carry byte
7. Add the first number (B) to accumulator (A)
8. If carry occurs -
→ Increment register D
9. Decrease register C by 1
10. Check whether $C = 0$
11. If not zero
Repeat steps 7-10
12. Check whether $C = 0$
13. If not zero
Repeat 7-10
14. Store accumulator's content in 2502H
15. Store reg. D content in 2503H
16. Stop Program.

Observation

Input

C050 = 02

C051 = 03

output

C060 = 06

Expt. No. _____

Page No. _____

30

Memorics

LXI H, C050H

MOV B, M

INX H

MOV C, M

MVI A, 00H

MVI

LOOP ADD B

DEC C

JNZ C008/LOOP

STA C060H

HLT

Result:- We successfully performed given
experiment multiplication of two 8 bit
numbers.

Teacher's Signature _____

Viva Question

- Q1 What is the function of XCHG instruction?
Ans It is used to exchange the data of the pair content with DE pair
- Q2 What is the function of JNZ instruction?
Ans Jump if not zero to the specified address if condition is true.
- Q3 What is the addressing mode of INX H?
Ans Register addressing mode.
- Q4 How many T states required to complete MOV A, 14?
Ans 7 T states.
- Q5 Define T State
Ans T-state is defined as one subdivision of the operation performed in one clock period. This is the unit in which, time of any operation is measured.